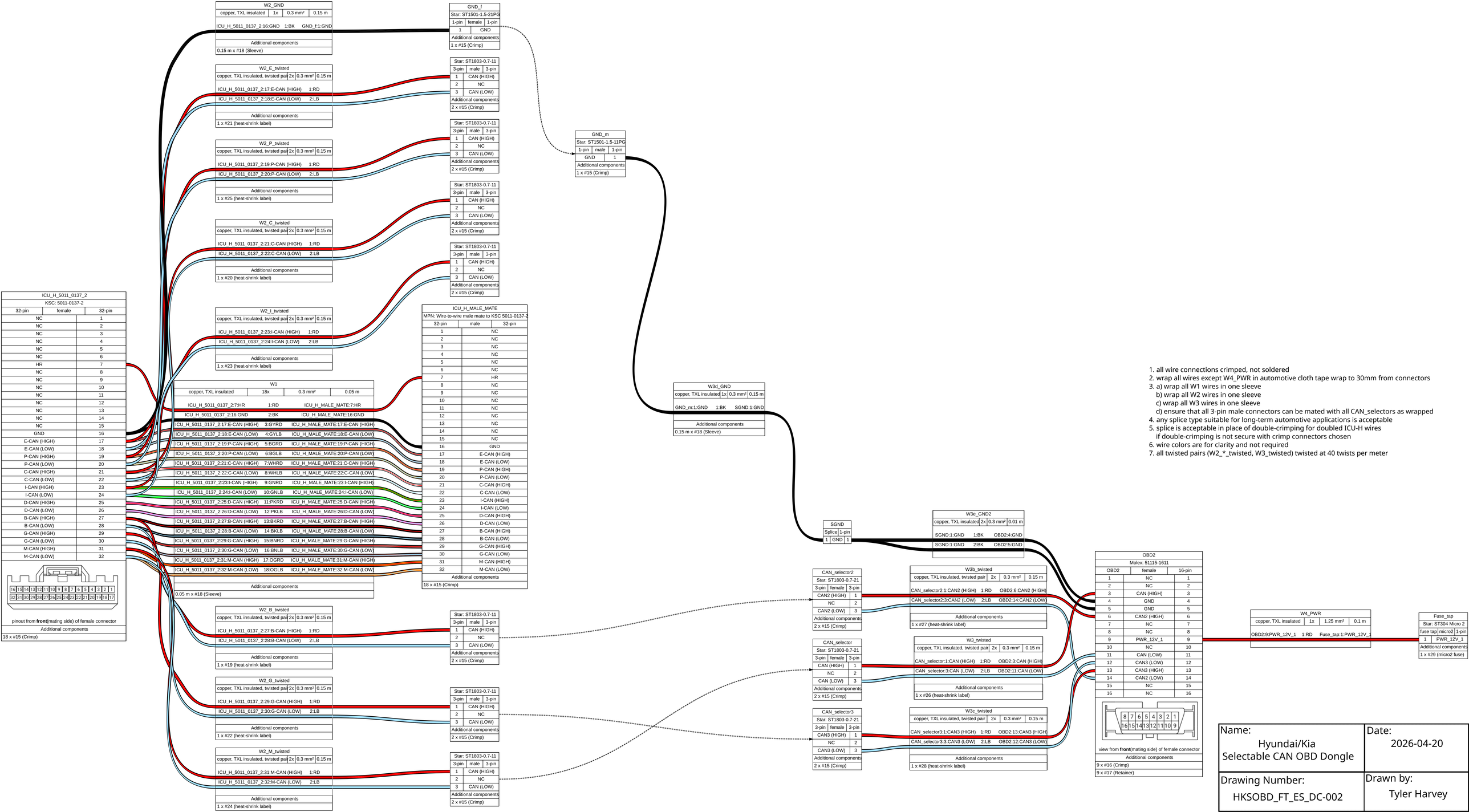
W2_GND
copper, TXL insulated 1x 0.3 mm² 0.15 m
ICU_H_5011_0137_2:16:GND 1:BK GND_f:1:GND
Additional components
0.15 m x #18 (Sleeve)
GND_f
Star: ST1501-1.5-21PG
1-pin female 1-pin
1 GND
Additional components
1 x #15 (Crimp)
W2_E_twisted
copper, TXL insulated, twisted pair 2x 0.3 mm² 0.15 m
ICU_H_5011_0137_2:17:E-CAN (HIGH) 1:RD
ICU_H_5011_0137_2:18:E-CAN (LOW) 2:LB
Additional components
1 x #21 (heat-shrink label)
Star: ST1803-0.7-11
3-pin male 3-pin
1 CAN (HIGH)
2 NC
3 CAN (LOW)
Additional components
2 x #15 (Crimp)
W2_P_twisted
copper, TXL insulated, twisted pair 2x 0.3 mm² 0.15 m
ICU_H_5011_0137_2:19:P-CAN (HIGH) 1:RD
ICU_H_5011_0137_2:20:P-CAN (LOW) 2:LB
Additional components
1 x #25 (heat-shrink label)
Star: ST1803-0.7-11
3-pin male 3-pin
1 CAN (HIGH)
2 NC
3 CAN (LOW)
Additional components
2 x #15 (Crimp)
GND_m
Star: ST1501-1.5-11PG
1-pin male 1-pin
GND 1
Additional components
1 x #15 (Crimp)
W2_C_twisted
copper, TXL insulated, twisted pair 2x 0.3 mm² 0.15 m
ICU_H_5011_0137_2:21:C-CAN (HIGH) 1:RD
ICU_H_5011_0137_2:22:C-CAN (LOW) 2:LB
Additional components
1 x #20 (heat-shrink label)
Star: ST1803-0.7-11
3-pin male 3-pin
1 CAN (HIGH)
2 NC
3 CAN (LOW)
Additional components
2 x #15 (Crimp)
Star: ST1803-0.7-11
3-pin male 3-pin
1 CAN (HIGH)
2 NC
3 CAN (LOW)
Additional components
2 x #15 (Crimp)
ICU_H_5011_0137_2
KSC: 5011-0137-2
32-pin female 32-pin
NC 1
NC 2
NC 3
NC 4
NC 5
NC 6
HR 7
NC 8
NC 9
NC 10
NC 11
NC 12
NC 13
NC 14
NC 15
GND 16
E-CAN (HIGH) 17
E-CAN (LOW) 18
P-CAN (HIGH) 19
P-CAN (LOW) 20
C-CAN (HIGH) 21
C-CAN (LOW) 22
I-CAN (HIGH) 23
I-CAN (LOW) 24
D-CAN (HIGH) 25
D-CAN (LOW) 26
B-CAN (HIGH) 27
B-CAN (LOW) 28
G-CAN (HIGH) 29
G-CAN (LOW) 30
M-CAN (HIGH) 31
M-CAN (LOW) 32
16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1
32 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17
pinout from front(mating side) of female connector
Additional components
18 x #15 (Crimp)
W2_I_twisted
copper, TXL insulated, twisted pair 2x 0.3 mm² 0.15 m
ICU_H_5011_0137_2:23:I-CAN (HIGH) 1:RD
ICU_H_5011_0137_2:24:I-CAN (LOW) 2:LB
Additional components
1 x #23 (heat-shrink label)
ICU_H_MALE_MATE
MPN: Wire-to-wire male mate to KSC 5011-0137-2
32-pin male 32-pin
1 NC
2 NC
3 NC
4 NC
5 NC
6 NC
7 HR
8 NC
9 NC
10 NC
11 NC
12 NC
13 NC
14 NC
15 NC
16 GND
17 E-CAN (HIGH)
18 E-CAN (LOW)
19 P-CAN (HIGH)
20 P-CAN (LOW)
21 C-CAN (HIGH)
22 C-CAN (LOW)
23 I-CAN (HIGH)
24 I-CAN (LOW)
25 D-CAN (HIGH)
26 D-CAN (LOW)
27 B-CAN (HIGH)
28 B-CAN (LOW)
29 G-CAN (HIGH)
30 G-CAN (LOW)
31 M-CAN (HIGH)
32 M-CAN (LOW)
Additional components
18 x #15 (Crimp)
W1
copper, TXL insulated 18x 0.3 mm² 0.05 m
ICU_H_5011_0137_2:7:HR 1:RD ICU_H_MALE_MATE:7:HR
ICU_H_5011_0137_2:16:GND 2:BK ICU_H_MALE_MATE:16:GND
ICU_H_5011_0137_2:17:E-CAN (HIGH) 3:GYRD ICU_H_MALE_MATE:17:E-CAN (HIGH)
ICU_H_5011_0137_2:18:E-CAN (LOW) 4:GYLB ICU_H_MALE_MATE:18:E-CAN (LOW)
ICU_H_5011_0137_2:19:P-CAN (HIGH) 5:BGRD ICU_H_MALE_MATE:19:P-CAN (HIGH)
ICU_H_5011_0137_2:20:P-CAN (LOW) 6:BGLB ICU_H_MALE_MATE:20:P-CAN (LOW)
ICU_H_5011_0137_2:21:C-CAN (HIGH) 7:WHRD ICU_H_MALE_MATE:21:C-CAN (HIGH)
ICU_H_5011_0137_2:22:C-CAN (LOW) 8:WHLB ICU_H_MALE_MATE:22:C-CAN (LOW)
ICU_H_5011_0137_2:23:I-CAN (HIGH) 9:GNRD ICU_H_MALE_MATE:23:I-CAN (HIGH)
ICU_H_5011_0137_2:24:I-CAN (LOW) 10:GNLB ICU_H_MALE_MATE:24:I-CAN (LOW)
ICU_H_5011_0137_2:25:D-CAN (HIGH) 11:PKRD ICU_H_MALE_MATE:25:D-CAN (HIGH)
ICU_H_5011_0137_2:26:D-CAN (LOW) 12:PKLB ICU_H_MALE_MATE:26:D-CAN (LOW)
ICU_H_5011_0137_2:27:B-CAN (HIGH) 13:BKRD ICU_H_MALE_MATE:27:B-CAN (HIGH)
ICU_H_5011_0137_2:28:B-CAN (LOW) 14:BKLB ICU_H_MALE_MATE:28:B-CAN (LOW)
ICU_H_5011_0137_2:29:G-CAN (HIGH) 15:BNRD ICU_H_MALE_MATE:29:G-CAN (HIGH)
ICU_H_5011_0137_2:30:G-CAN (LOW) 16:BNLB ICU_H_MALE_MATE:30:G-CAN (LOW)
ICU_H_5011_0137_2:31:M-CAN (HIGH) 17:OGRD ICU_H_MALE_MATE:31:M-CAN (HIGH)
ICU_H_5011_0137_2:32:M-CAN (LOW) 18:OGLB ICU_H_MALE_MATE:32:M-CAN (LOW)
Additional components
0.05 m x #18 (Sleeve)
W3d_GND
copper, TXL insulated 1x 0.3 mm² 0.15 m
GND_m:1:GND 1:BK SGND:1:GND
Additional components
0.15 m x #18 (Sleeve)
SGND
Splice 1-pin
1 GND 1
W3e_GND2
copper, TXL insulated 2x 0.3 mm² 0.01 m
SGND:1:GND 1:BK OBD2:4:GND
SGND:1:GND 2:BK OBD2:5:GND
W2_B_twisted
copper, TXL insulated, twisted pair 2x 0.3 mm² 0.15 m
ICU_H_5011_0137_2:27:B-CAN (HIGH) 1:RD
ICU_H_5011_0137_2:28:B-CAN (LOW) 2:LB
Additional components
1 x #19 (heat-shrink label)
Star: ST1803-0.7-11
3-pin male 3-pin
1 CAN (HIGH)
2 NC
3 CAN (LOW)
Additional components
2 x #15 (Crimp)
CAN_selector2
Star: ST1803-0.7-21
3-pin female 3-pin
CAN2 (HIGH) 1
NC 2
CAN2 (LOW) 3
Additional components
2 x #15 (Crimp)
W3b_twisted
copper, TXL insulated, twisted pair 2x 0.3 mm² 0.15 m
CAN_selector2:1:CAN2 (HIGH) 1:RD OBD2:6:CAN2 (HIGH)
CAN_selector2:3:CAN2 (LOW) 2:LB OBD2:14:CAN2 (LOW)
Additional components
1 x #27 (heat-shrink label)
W2_G_twisted
copper, TXL insulated, twisted pair 2x 0.3 mm² 0.15 m
ICU_H_5011_0137_2:29:G-CAN (HIGH) 1:RD
ICU_H_5011_0137_2:30:G-CAN (LOW) 2:LB
Additional components
1 x #22 (heat-shrink label)
Star: ST1803-0.7-11
3-pin male 3-pin
1 CAN (HIGH)
2 NC
3 CAN (LOW)
Additional components
2 x #15 (Crimp)
CAN_selector
Star: ST1803-0.7-21
3-pin female 3-pin
CAN (HIGH) 1
NC 2
CAN (LOW) 3
Additional components
2 x #15 (Crimp)
W3_twisted
copper, TXL insulated, twisted pair 2x 0.3 mm² 0.15 m
CAN_selector:1:CAN (HIGH) 1:RD OBD2:3:CAN (HIGH)
CAN_selector:3:CAN (LOW) 2:LB OBD2:11:CAN (LOW)
Additional components
1 x #26 (heat-shrink label)
W2_M_twisted
copper, TXL insulated, twisted pair 2x 0.3 mm² 0.15 m
ICU_H_5011_0137_2:31:M-CAN (HIGH) 1:RD
ICU_H_5011_0137_2:32:M-CAN (LOW) 2:LB
Additional components
1 x #24 (heat-shrink label)
Star: ST1803-0.7-11
3-pin male 3-pin
1 CAN (HIGH)
2 NC
3 CAN (LOW)
Additional components
2 x #15 (Crimp)
CAN_selector3
Star: ST1803-0.7-21
3-pin female 3-pin
CAN3 (HIGH) 1
NC 2
CAN3 (LOW) 3
Additional components
2 x #15 (Crimp)
W3c_twisted
copper, TXL insulated, twisted pair 2x 0.3 mm² 0.15 m
CAN_selector3:1:CAN3 (HIGH) 1:RD OBD2:13:CAN3 (HIGH)
CAN_selector3:3:CAN3 (LOW) 2:LB OBD2:12:CAN3 (LOW)
Additional components
1 x #28 (heat-shrink label)
OBD2
Molex: 51115-1611
OBD2 female 16-pin
1 NC 1
2 NC 2
3 CAN (HIGH) 3
4 GND 4
5 GND 5
6 CAN2 (HIGH) 6
7 NC 7
8 NC 8
9 PWR_12V_1 9
10 NC 10
11 CAN (LOW) 11
12 CAN3 (LOW) 12
13 CAN3 (HIGH) 13
14 CAN2 (LOW) 14
15 NC 15
16 NC 16
8 7 6 5 4 3 2 1
16 15 14 13 12 11 10 9
view from front(mating side) of female connector
Additional components
9 x #16 (Crimp)
9 x #17 (Retainer)
W4_PWR
copper, TXL insulated 1x 1.25 mm² 0.1 m
OBD2:9:PWR_12V_1 1:RD Fuse_tap:1:PWR_12V_1
Fuse_tap
Star: ST304 Micro 2
fuse tap micro2 1-pin
1 PWR_12V_1
Additional components
1 x #29 (micro2 fuse)
1. all wire connections crimped, not soldered
2. wrap all wires except W4_PWR in automotive cloth tape wrap to 30mm from connectors
3. a) wrap all W1 wires in one sleeve
b) wrap all W2 wires in one sleeve
c) wrap all W3 wires in one sleeve
d) ensure that all 3-pin male connectors can be mated with all CAN_selectors as wrapped
4. any splice type suitable for long-term automotive applications is acceptable
5. splice is acceptable in place of double-crimping for doubled ICU-H wires
if double-crimping is not secure with crimp connectors chosen
6. wire colors are for clarity and not required
7. all twisted pairs (W2_*_twisted, W3_twisted) twisted at 40 twists per meter
Name:
Hyundai/Kia
Selectable CAN OBD Dongle
Date:
2026-04-20
Drawing Number:
HKSOBD_FT_ES_DC-002
Drawn by:
Tyler Harvey